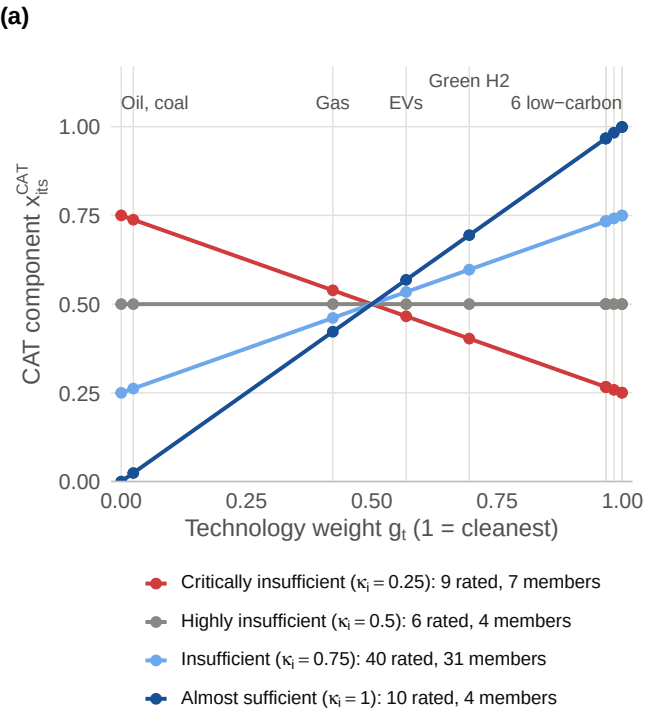
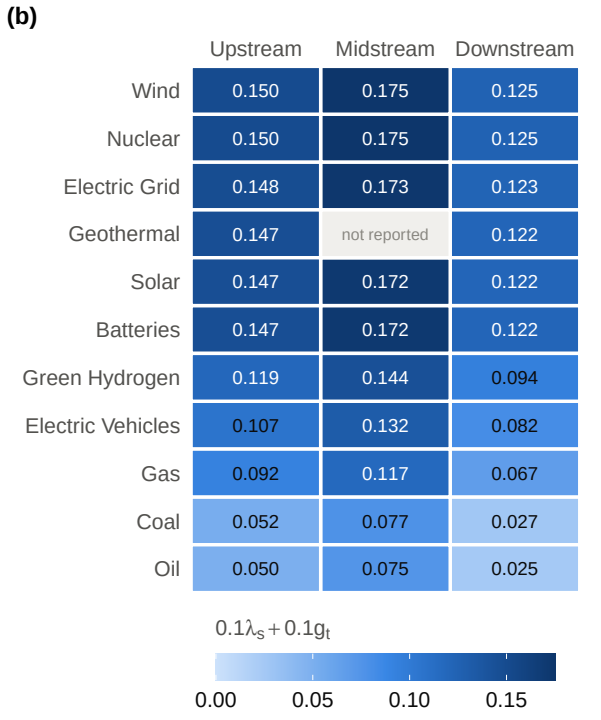

(a)
Green H2
Oil, coal
Gas
EVs
6 low−carbon
1.00
0.75
0.50
0.25
0.00
CAT component x_{its}^{CAT}
0.00
0.25
0.50
0.75
1.00
Technology weight g_t (1 = cleanest)
Critically insufficient ($\kappa_i = 0.25$): 9 rated, 7 members
Highly insufficient ($\kappa_i = 0.5$): 6 rated, 4 members
Insufficient ($\kappa_i = 0.75$): 40 rated, 31 members
Almost sufficient ($\kappa_i = 1$): 10 rated, 4 members


(b)
Upstream
Midstream
Downstream
Wind 0.150 0.175 0.125
Nuclear 0.150 0.175 0.125
Electric Grid 0.148 0.173 0.123
Geothermal 0.147 not reported 0.122
Solar 0.147 0.172 0.122
Batteries 0.147 0.172 0.122
Green Hydrogen 0.119 0.144 0.094
Electric Vehicles 0.107 0.132 0.082
Gas 0.092 0.117 0.067
Coal 0.052 0.077 0.027
Oil 0.050 0.075 0.025
$0.1\lambda_s + 0.1g_t$
0.00
0.05
0.10
0.15